

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc Biotechnology Examination April-May 2018

Course code & Name: BT853 Cancer Biology

04.05.2018

Friday

10:00 AM To 10:30 AM

Maximum Marks: 20

MCQ

Important Instructions:

- Tick the correct answer and it should be written in question paper itself.
- Use of non-programmable calculator is allowed.

Q - I Choose the correct answer for the following questions.

20

1. X-chromosome inactivation occurs due to
 - (i) miRNAs deregulation
 - (ii) DNA methylation
 - (iii) Histone acetylation
 - (iv) Nucleosome positioning
2. RAS is a
 - (i) Transmembrane receptor
 - (ii) Cytosolic signaling molecule
 - (iii) Ligand
 - (iv) Transcription factor
3. Cancer with highest hereditary predisposition is
 - (i) Lung cancer
 - (ii) Prostate cancer
 - (iii) Ovarian cancer
 - (iv) Breast cancer
4. Proto-oncogene becomes oncogene due to
 - (i) Gene amplification
 - (ii) Point Mutations
 - (iii) Indels
 - (iv) All the above
5. miRNA is made up of approximately _____ nucleotides
 - (i) 10
 - (ii) 20
 - (iii) 30
 - (iv) 40

6. Approximate number of cancer cases diagnosed globally per year are
- | | |
|--------------------------------|---------------------------------|
| (i) 2 million to 4 million | (ii) 1.2 million to 1.4 million |
| (iii) 12 million to 14 million | (iv) 120 million to 140 million |
7. Colon cancer risk is reduced by high intake of
- | | |
|----------------------|-------------------|
| (i) Leafy vegetables | (ii) Tea |
| (iii) Red Meat | (iv) Fruit juices |
8. IARC is located in
- | | |
|--------------|-------------|
| (i) Germany | (ii) France |
| (iii) Canada | (iv) Japan |
9. Which of the following is least common cancer among men
- | | |
|-------------------|----------------------|
| (i) Testis cancer | (ii) Prostate cancer |
| (iii) Lung cancer | (iv) Stomach cancer |
10. Select the odd one out
- | | |
|------------|-----------|
| (i) HRAS | (ii) KRAS |
| (iii) NRAS | (iv) PRAS |
11. Proto-oncogenes are
- | | |
|-----------------------------------|-------------------------------|
| (i) Apoptosis promoting genes | (iv) Growth promoting genes |
| (iii) Apoptosis suppressing genes | (ii) Growth suppressing genes |
12. Regular exposure of ER of _____ wavelength may cause permanent DNA damage
- | | |
|---------------------|--------------------|
| (i) 1000 to 2000 nm | (ii) 300 to 700 nm |
| (iii) 100 to 200 nm | (iv) All the above |
13. Select the odd cancer type
- | | |
|-----------------------------|-------------------------|
| (i) Epididymal cancer | (ii) Endometrial cancer |
| (iii) Fallopian tube cancer | (iv) Vulvar cancer |
14. Expand TNM _____
15. Biological damage caused by radiation is measured in _____ units
16. Expand MCPyV _____
17. Cholangiocarcinoma is cancer of _____
18. Infection of HBV causes bladder cancer and infection of HCV causes Cholangiocarcinoma
- | | |
|------|-------|
| True | False |
|------|-------|
19. Persistent inflammation may cause cancer
- | | |
|------|-------|
| True | False |
|------|-------|
20. Mutation(s) in tumor suppressor genes is the only factor of carcinogenesis
- | | |
|------|-------|
| True | False |
|------|-------|

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

IV Semester of MSc Biotechnology Examination April-May 2018

Course code & Name: BT853 Cancer Biology

04.05.2018

Friday

10:31 AM To 01:00 PM

Maximum Marks: 50

Instructions:

1. Section I and II must be attempted in TWO ANSWER SHEET.
2. Make suitable assumptions and draw neat figures wherever required.
3. Use of non-programmable calculator is allowed.
4. Show necessary calculations.

SECTION -I

Q - II Attempt any FOUR

20

1. How infection of certain pathogens can cause cancer. Discuss in detail
2. What is the role of IARC, France and NCRP, India in relation to cancer epidemiology?
3. Discuss various cancer etiologic agents in brief.
4. Discuss tumor staging and grading in cancer?
5. What is hereditary cancer? Describe its various characteristics.

SECTION – II

Q-III Attempt any SIX

30

1. What are the hallmarks of cancer? Describe any one in detail.
2. Describe dual role of p53 and MYC in cancer
3. Discuss cancer prevention
4. How miRNAs exhibit both tumor suppressor as well as oncogenic effects and how epigenetics can be helpful in the management of cancer
5. Discuss RAS-MAPK pathway and its importance in carcinogenesis
6. Discuss “CAUTION” in cancer
7. What is two-hit hypothesis and how tumor suppressor and oncogenes causes cancer?